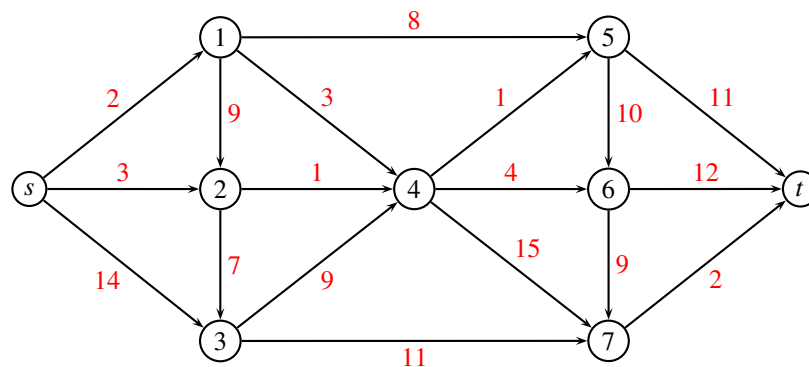


Take-home exam: December 18, 2013.

Course name: Network Optimization

Course no: 42115

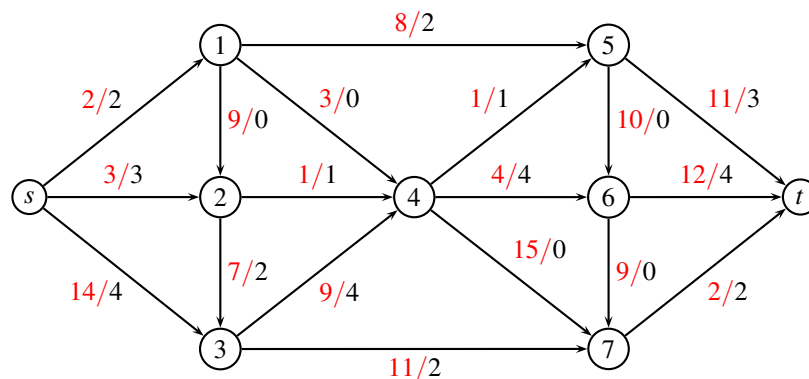
Maximum Flow



Answer 1 We have the following iterations

- $s346t$, 4 units
- $s15t$, 2 units
- $s237t$, 2 units
- $s245t$, 1 unit

This results in the following solution

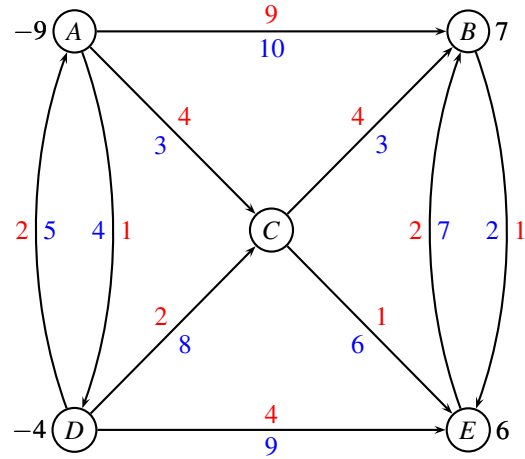


■

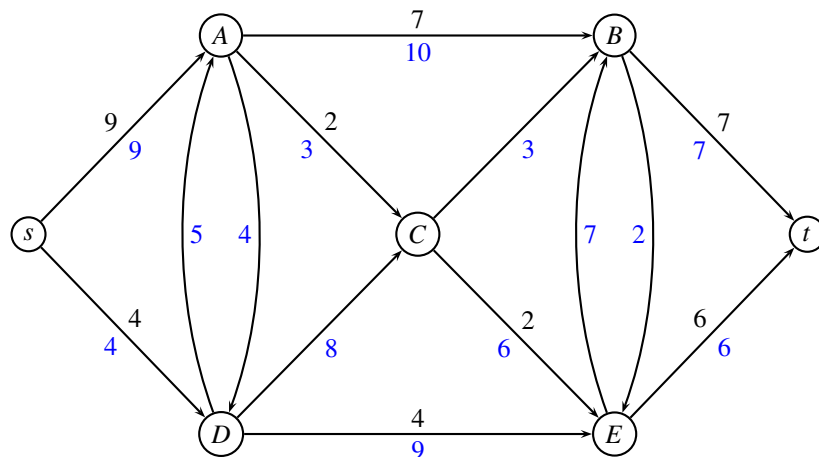
Answer 2 A minimum cut is $S = \{s, 2, 3, 4, 7\}$ and $T = \{1, 5, 6, t\}$ with capacity $2 + 1 + 4 + 2 = 9$. The value of the flow is also 9. ■

Min cost flow

We have the problem

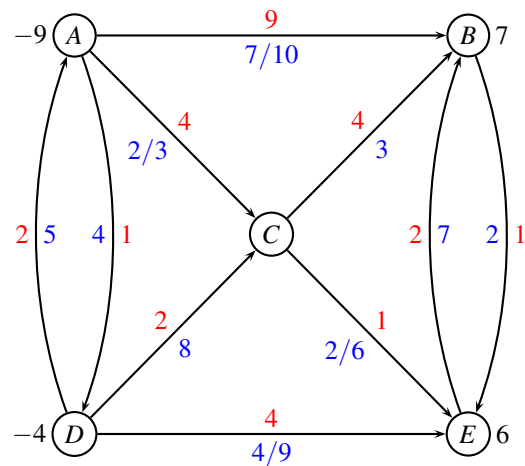


Answer 3 We introduce two extra nodes s and t . Edges are added from s to all nodes with $b_i < 0$, and from all nodes with $b_i > 0$ to t . Capacity of edges is set to $|b_i|$. All costs are ignored.



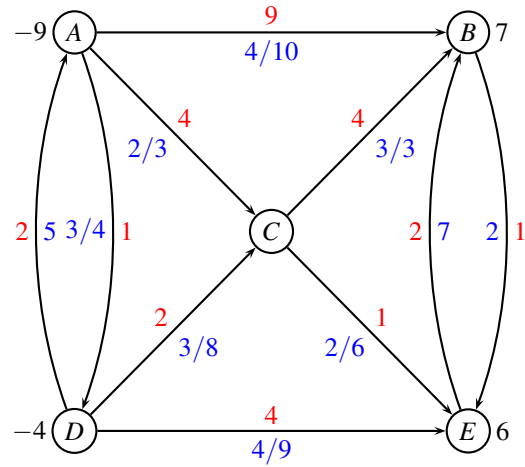
A feasible flow is indicated with black numbers. ■

Answer 4 We have the initial solution

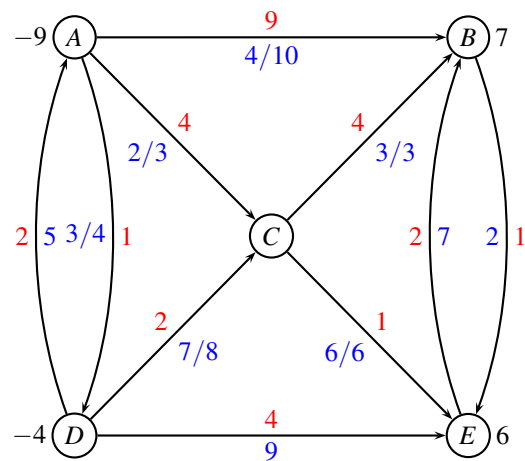


with objective $7 \cdot 9 + 2 \cdot 4 + 2 \cdot 1 + 4 \cdot 4 = 89$

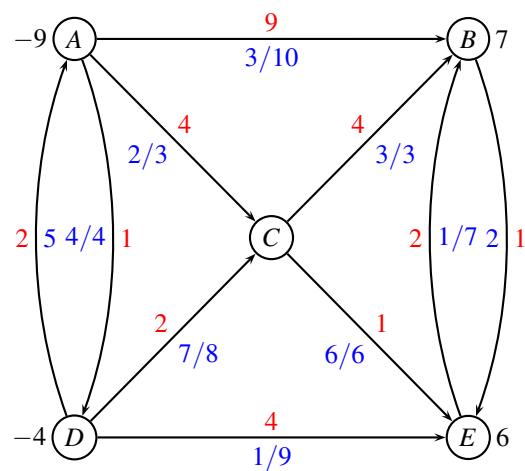
Finding augmenting cycle A, D, C, B, A with cost $1 + 2 + 4 - 9 = -2$. Flowing 3 units.



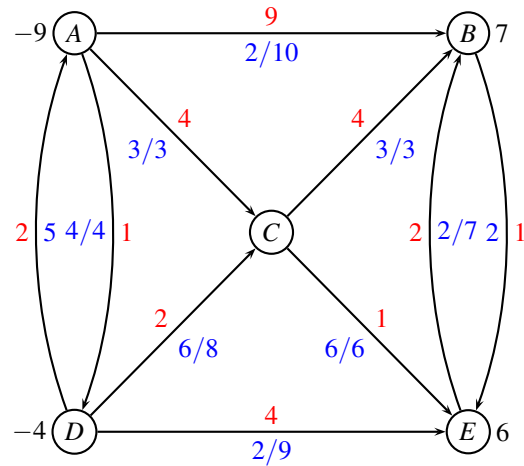
Finding augmenting cycle D, C, E, D with cost $2 + 1 - 4 = -1$. Flowing 4 units.



Finding augmenting cycle A, D, E, B, A with cost $1 + 4 + 2 - 9 = -2$. Flowing 1 unit.



Finding augmenting cycle A, C, D, E, B, A with cost $4 - 2 + 4 + 2 - 9 = -1$. Flowing 1 unit.



No more negative cycles. The optimal solution value is $z = 76$.

Checking solution with CPLEX:

```

minimize
  9xAB+4xAC+1xAD+1xBE+4xCB+1xCE+2xDA+2xDC+4xDE+2xEB

subject to
  -xAB-xAC-xAD+xDA=-9
  xAB-xBE+xCB+xEB=7
  xAC-xCB-xCE+xDC=0
  xAD-xDA-xDC-xDE=-4
  xBE+xCE+xDE-xEB=6

  xAB <= 10
  xAC <= 3
  xAD <= 4
  xBE <= 2
  xCB <= 3
  xCE <= 6
  xDA <= 5
  xDC <= 8
  xDE <= 9
  xEB <= 7
end

CPLEX> read mcf
Problem 'mcf.lp' read.
Read time = 0.02 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 10 rows and 0 columns.
Aggregator did 1 substitutions.
Reduced LP has 4 rows, 9 columns, and 18 nonzeros.
Presolve time = 0.00 sec.

Iteration log . . .
Iteration: 1 Dual objective = 50.000000

Dual simplex - Optimal: Objective = 7.6000000000e+01
Solution time = 0.00 sec. Iterations = 3 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
xAB                2.000000

```

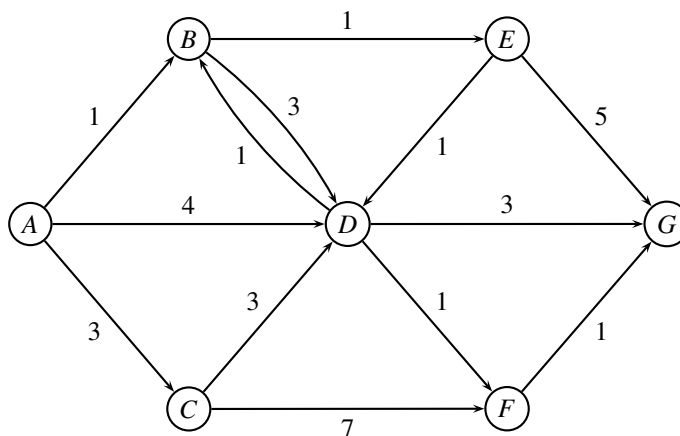
```

xAC          3.000000
xAD          4.000000
xCB          3.000000
xCE          6.000000
xDC          6.000000
xDE          2.000000
xEB          2.000000
All other variables in the range 1-10 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
c1                  -7.000000
c2                   2.000000
c3                  -2.000000
c4                  -4.000000
c7                  -1.000000
c8                  -2.000000
c11                 -1.000000
All other dual prices in the range 1-15 are 0.

```

■

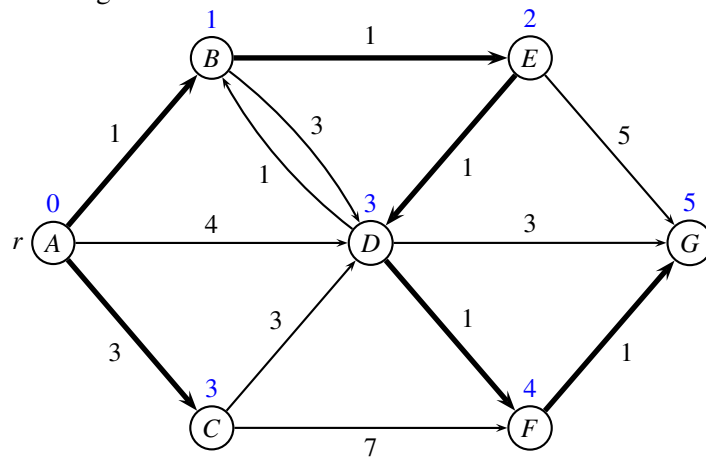
Shortest path



Answer 5 Running Dijkstra we have the following potentials in each iteration:

relaxing	potential						
	A	B	C	D	E	F	G
init	0	∞	∞	∞	∞	∞	∞
A		1	3	4			
B					2		
E				3			7
C						10	
D						4	6
F							5
G							

This results in the following tree



■

Answer 6 The travel times t are

	distance						
	A	B	C	D	E	F	G
		1	1	3	2	4	5

■

Answer 7 We use the label correcting algorithm as follows:

- We start with the following labels:

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5

and we have the queue $Q = \{A, B, C, D, E, F, G\}$.

- Extract A from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1						

and we have the queue $Q = \{B, C, D, E, F, G\}$.

- Extract B from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1						
						4	2						

The label (4,2) for D is dominated, so we remove it. The queue is $Q = \{C, D, E, F, G\}$.

- Extract C from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1						
						6	2						

The label (6,2) for D is dominated, so we remove it. The queue is $Q = \{D, E, F, G\}$.

- Extract D from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
		4	4			4	1			5	2	6	4
		5	2									7	2

The labels (4,4) and (5,2) for B are dominated so we remove them. This means that no labels for B has been changed, and we do not need to add B to the queue. Labels for F and G have been modified, but the nodes are already in the queue. The queue is $Q = \{E, F, G\}$.

- Extract E from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1			5	2	6	4
												7	2
												7	3

The label (7,3) for G is dominated so we remove it. The queue is $Q = \{F, G\}$.

- Extract F from the queue, and update all outgoing edges

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1			5	2	6	4
												7	2
												6	3

The label (6,4) for G is dominated so we remove it. The queue is $Q = \{G\}$.

- Extract G from the queue. No outgoing edges from G .

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	5	5
						4	1			5	2	6	3
										7	2		

The queue is empty, we stop.

We were asked to find the shortest distance to every node using at most 4 time units. Hence the cheapest cost with $t \leq 4$ in each column

A		B		C		D		E		F		G	
w	t	w	t	w	t	w	t	w	t	w	t	w	t
0	0	1	1	3	1	3	3	2	2	4	4	6	3

■

Nurse scheduling

Answer 8 We have the requirements

shift i	1	2	3	4	5	6
time	6:00-10:00	10:00-14:00	14:00-18:00	18:00-22:00	22:00-2:00	2:00-6:00
min nurses d_i	70	80	50	60	40	30

First we notice that we can have the following duties:

duty j	start a_j	end b_j	cost c_j
1	1	2	2
2	2	3	2
3	3	4	2
4	4	5	3
5	5	6	4
6	6	6	2

Writing up the constraints in an IP model, where x_i denotes the number of nurses following duty i , we get

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} \geq \begin{pmatrix} 70 \\ 80 \\ 50 \\ 60 \\ 40 \\ 30 \end{pmatrix}$$

Since the model has consecutive 1's in every column, we can use the modeling technique from Application 9.6 in Ahuja, Orlin, Magnanti. First, we add surplus variables to each row to get equations, and

then we add an extra zero-row.

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 70 \\ 80 \\ 50 \\ 60 \\ 40 \\ 30 \\ 0 \end{pmatrix}$$

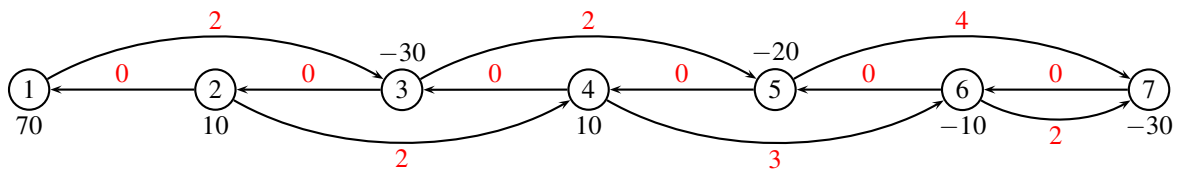
Using row-operations we can get it to the form

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 70 \\ 10 \\ -30 \\ 10 \\ -20 \\ -10 \\ -30 \end{pmatrix}$$

This can be recognised as a min-cost-flow problem.

To be more specific, we construct a node for every shift $1, \dots, 6$ and an additional end node 7. The demand b_i in node i is $-(d_i - d_{i-1})$. (Negative values indicate sinks, positive values sources).

For every duty j we introduce an edge from node a_j to $b_j + 1$ having cost c_j . The flow on this edge indicates how many nurses follow duty j . Moreover, we introduce edges from node $i + 1$ to i having cost 0. These edges allow for "over-covering" shifts, i.e. nurses can end later than we actually need them. The flow on these edges corresponds to the surplus variables y_i . The capacity of all edges is set to ∞ .



■

Answer 9 We get the LP-model

minimize $2x_{13} + 2x_{24} + 2x_{35} + 3x_{46} + 4x_{57} + 2x_{67}$

subject to
 $x_{13} - x_{21} = 70$ \\ flow balance node 1
 $x_{24} + x_{21} - x_{32} = 10$ \\ flow balance node 2
 $x_{35} + x_{32} - x_{43} - x_{13} = -30$ \\ flow balance node 3
 $x_{46} + x_{43} - x_{54} - x_{24} = 10$ \\ flow balance node 4
 $x_{57} + x_{54} - x_{65} - x_{35} = -20$ \\ flow balance node 5
 $x_{67} + x_{65} - x_{76} - x_{46} = -10$ \\ flow balance node 6
 $+x_{76} - x_{57} - x_{67} = -30$ \\ flow balance node 7
end

■

Answer 10 The optimal solution is $x_{13} = 70$, $x_{24} = 10$, $x_{35} = 40$, $x_{57} = 20$, $x_{43} = 20$, $x_{76} = 10$. So,

duty j	start a_j	end b_j	cost c_j	nurses
1	1	2	2	70
2	2	3	2	10
3	3	4	2	40
4	4	5	3	20
5	5	6	4	20
6	6	6	2	10

No "over-covering" is necessary since no "backward" edges are used. The overall cost is $z = 400$.

```
CPLEX> read nurses
Problem 'nurses.lp' read.
Read time =    0.04 sec.
CPLEX> opt
Tried aggregator 1 time.
Aggregator did 1 substitutions.
Reduced LP has 6 rows, 11 columns, and 22 nonzeros.
Presolve time =    0.00 sec.
Initializing dual steep norms . . .

Iteration log . . .
Iteration:    1   Dual objective      =          260.000000

Dual simplex - Optimal:  Objective =  4.000000000000e+02
Solution time =    0.02 sec.  Iterations = 5 (0)

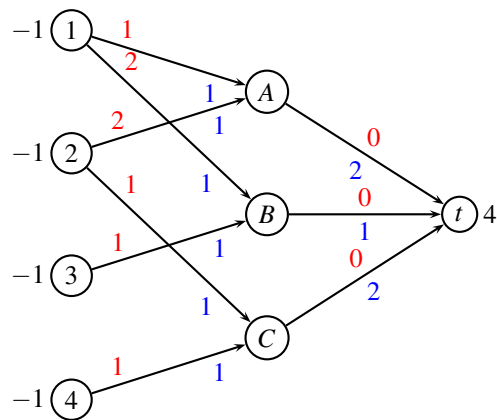
CPLEX> dis sol var -
Variable Name      Solution Value
x13                70.000000
x24                10.000000
x35                40.000000
x46                20.000000
x57                20.000000
x67                10.000000
All other variables in the range 1-12 are 0.
```

■

Student assignment

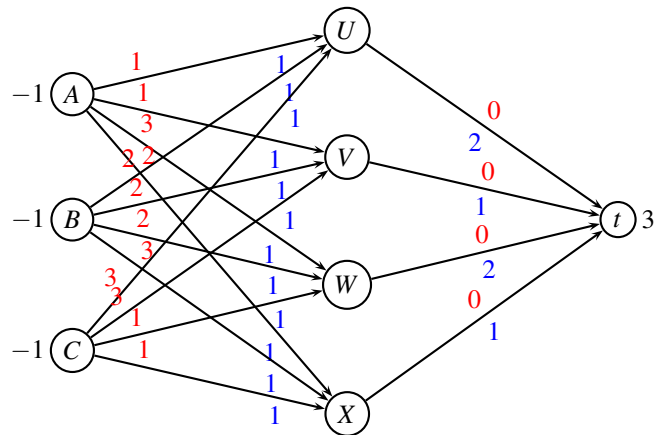
Answer 11 The problem can be solved as two min-cost flow problems. First we assign students to projects. For every student we create a node, and for every project we create a node. The edge between a student node and a project node gets a cost corresponding to the priority, and a capacity equal to 1. An additional sink t is introduced. Every edge from a project node to t gets a capacity corresponding to the capacity of the project, and a cost of 0. Every student node gets demand -1 and t gets demand

equal to the sum of students.



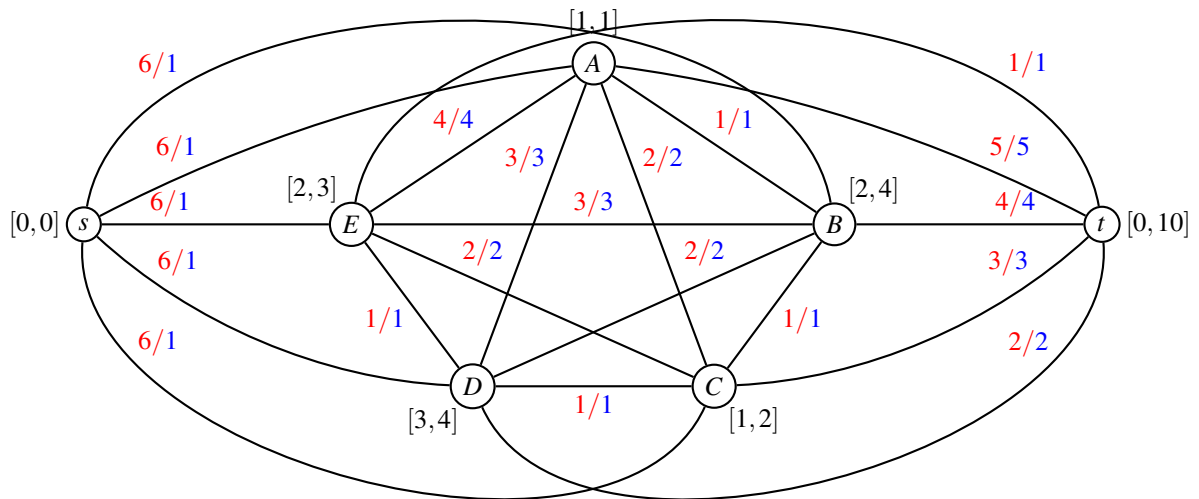
Red values indicate the cost, blue values the capacity of an edge. The node potentials are negative for a source, and positive for a sink.

Next, we assign projects to supervisors. As before we introduce a node for every project and a node for every professor. Edges between student nodes and professor nodes get cost equal to the preference, and capacity equal to 1. An additional sink t is introduced. Every edge from a professor node to t gets a capacity corresponding to the capacity of the professor, and a cost of 0. Every project node gets demand -1 and t gets demand equal to the sum of projects.



■

Vehicle Routing Problem with time constraints



Answer 12 A possible solution is

- Vehicle 1: $sABDt$
- Vehicle 2: $sCEt$

The overall cost is 20 ■

Answer 13 We have the route costs:

route	nodes	cost
1	s,A,B,D,t	11
2	s,B,D,t	10
3	s,C,B,D,t	11
4	s,C,D,t	9
5	s,C,E,t	9
6	s,C,E,D,t	11
7	s,D,t	8
8	s,E,D,t	9
9	s,E,t	7

■

Answer 14 Waiting times are indicated in the below table. Each (wait) indicates one time unit.

route	nodes	wait
1	s,A,B,D,t	0
2	s,(wait),B,D,t	1
3	s,C,B,D,t	0
4	s,C,(wait),D,t	1
5	s,C,E,t	0
6	s,C,E,D,t	0
7	s,(wait),(wait),D,t	2
8	s,(wait),E,D,t	1
9	s,(wait),E,t	1

■

Answer 15 Since every customer must be visited at least once, we use a set-covering formulation. The problem with all 9 routes becomes:

```

minimize
    11x1+10x2+11x3+ 9x4+ 9x5+11x6+ 8x7+ 9x8+ 7x9

subject to
c1:  +x1                                     >= 1
c2:  +x1 +x2 +x3                             >= 1
c3:           +x3 +x4 +x5 +x6                 >= 1
c4:  +x1 +x2 +x3 +x4             +x6 +x7 +x8   >= 1
c5:           +x5 +x6             +x8 +x9 >= 1
end

CPLEX> read vrp
Problem 'vrp.lp' read.
Read time =    0.05 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 8 rows and 11 columns.
Reduced LP has 2 rows, 3 columns, and 4 nonzeros.
Presolve time =    0.00 sec.

Iteration log . . .
Iteration:    1    Dual objective      =          20.000000

Dual simplex - Optimal:  Objective =  2.0000000000e+01
Solution time =    0.02 sec.  Iterations = 2 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
x5                  1.000000
All other variables in the range 1-14 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
c1                   11.000000
c3                   9.000000

```

All other dual prices in the range 1-10 are 0.

The IP-optimal solution gives the solution value $z_{IP} = 20$. Routes 1 and 5 are chosen. ■

Answer 16 We run column generation as follows:

- **Initial Solution.** We can choose the trivial routes:

tour	cost
0,A,6	11
0,B,6	10
0,C,6	9
0,D,6	8
0,E,6	7

which gives a solution of 45.

- **Column generation.** We start with formulation

$$\begin{array}{llll}
 \min & 11x_a + 10x_b + 9x_c + 8x_d + 7x_e \\
 \text{s.t.} & x_a & & & \geq 1 \\
 & & x_b & & \geq 1 \\
 & & & x_c & \geq 1 \\
 & & & & x_d & \geq 1 \\
 & & & & & x_e & \geq 1
 \end{array}$$

the dual variables corresponding to the constraints are

$$y_a = 11, y_b = 10, y_c = 9, y_d = 8, y_e = 7$$

The reduced cost of a route is now the “original” cost of the edges minus the dual variables corresponding to the customers visited.

route	nodes	original cost	dual	reduced cost
1	s,A,B,D,t	11	11+10+8	-18
2	s,B,D,t	10	10+8	-8
3	s,C,B,D,t	11	10+9+8	-16
4	s,C,D,t	9	9+8	-8
5	s,C,E,t	9	9+7	-7
6	s,C,E,D,t	11	9+8+7	-13
7	s,D,t	8	8	0
8	s,E,D,t	9	8+7	-6
9	s,E,t	7	7	0

The most beneficial route is route 1, i.e. (s, A, B, D, t) having the cost $11 - (11 + 10 + 8) = -18$. Since the reduced cost of the new column is negative, we add it to the problem, getting:

$$\begin{array}{llll}
 \min & 11x_a + 10x_b + 9x_c + 8x_d + 7x_e + 11x_1 \\
 \text{s.t.} & x_a & & & +x_1 & \geq 1 \\
 & & x_b & & +x_1 & \geq 1 \\
 & & & x_c & & \geq 1 \\
 & & & & x_e & +x_1 & \geq 1 \\
 & & & & & x_e & \geq 1
 \end{array}$$

the dual variables are

$$y_a = 11, y_b = 0, y_c = 9, y_d = 0, y_e = 7$$

The reduced costs are

route	nodes	original cost	dual	reduced cost
1	s,A,B,D,t	11	11+0+0	0
2	s,B,D,t	10	0+0	10
3	s,C,B,D,t	11	0+9+0	2
4	s,C,D,t	9	9+0	0
5	s,C,E,t	9	9+7	-7
6	s,C,E,D,t	11	9+0+7	-5
7	s,D,t	8	8	0
8	s,E,D,t	9	0+7	2
9	s,E,t	7	7	0

The most beneficial route is now route 5 i.e. (s, C, E, t) having the reduced cost $9 - (9 + 7) = -7$, which is added to the problem:

$$\begin{array}{llllll}
 \min & 11x_a + 10x_b + 9x_c + 8x_d + 7x_e + 11x_1 + 9x_5 \\
 \text{s.t.} & x_a & & & & +x_1 & \geq 1 \\
 & & x_b & & & +x_1 & \geq 1 \\
 & & & x_c & & & +x_5 \geq 1 \\
 & & & & x_d & +x_1 & \geq 1 \\
 & & & & & x_e & +x_5 \geq 1
 \end{array}$$

the dual variables are

$$y_a = 11, y_b = 0, y_c = 9, y_d = 0, y_e = 0$$

Reduced costs are

route	nodes	original cost	dual	reduced cost
1	s,A,B,D,t	11	11+0+0	0
2	s,B,D,t	10	0+0	10
3	s,C,B,D,t	11	0+9+0	2
4	s,C,D,t	9	9+0	0
5	s,C,E,t	9	9+0	0
6	s,C,E,D,t	11	9+0+0	2
7	s,D,t	8	8	0
8	s,E,D,t	9	0+0	9
9	s,E,t	7	0	0

Now, all routes have reduced costs ≥ 0 , so we are done. The bound found by column generation is $z_{LP} = 20$.

■

Answer 17 Compare bounds. We have $z_{IP} = z_{LP}$ so the bounds are tight. Actually the LP-solution returned an integer solution, and hence solved the problem. ■

Total model

```
minimize
    11x1+10x2+11x3+ 9x4+ 9x5+11x6+ 8x7+ 9x8+ 7x9+11xa+10xb+ 9xc+ 8xd+ 7xe

subject to
c1:  +x1                                     +xa                                     >= 1
c2:  +x1  +x2  +x3                                     +xb                                     >= 1
c3:                                     +x3  +x4  +x5  +x6                                     +xc                                     >= 1
c4:  +x1  +x2  +x3  +x4                                     +x6  +x7  +x8                                     +xd                                     >= 1
c5:                                     +x5  +x6                                     +x8  +x9                                     +xe >= 1

\\ r1: x1 <= 0
\\ r2: x2 <= 0
\\ r3: x3 <= 0
\\ r4: x4 <= 0
\\ r5: x5 <= 0
\\ r6: x6 <= 0
\\ r7: x7 <= 0
\\ r8: x8 <= 0
\\ r9: x9 <= 0

s1: xa <= 0
s2: xb <= 0
s3: xc <= 0
s4: xd <= 0
s5: xe <= 0

end

CPLEX> read vrp
Problem 'vrp.lp' read.
Read time = 0.05 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 8 rows and 11 columns.
Reduced LP has 2 rows, 3 columns, and 4 nonzeros.
Presolve time = 0.00 sec.

Iteration log . . .
Iteration: 1 Dual objective = 20.000000

Dual simplex - Optimal: Objective = 2.0000000000e+01
Solution time = 0.02 sec. Iterations = 2 (0)

CPLEX> dis sol var -
Variable Name          Solution Value
x1                      1.000000
x5                      1.000000
All other variables in the range 1-14 are 0.
CPLEX> dis sol dua -
Constraint Name          Dual Price
c1                      11.000000
c3                      9.000000
All other dual prices in the range 1-10 are 0.
```

Iteration 0

```
minimize
    11x1+10x2+11x3+ 9x4+ 9x5+11x6+ 8x7+ 9x8+ 7x9+11xa+10xb+ 9xc+ 8xd+ 7xe

subject to
c1:  +x1                                +xa                                >= 1
c2:  +x1  +x2  +x3                                +xb                                >= 1
c3:                                +x3  +x4  +x5  +x6                                +xc                                >= 1
c4:  +x1  +x2  +x3  +x4                                +x6  +x7  +x8                                +xd                                >= 1
c5:                                +x5  +x6                                +x8  +x9                                +xe >= 1

r1: x1 <= 0
r2: x2 <= 0
r3: x3 <= 0
r4: x4 <= 0
r5: x5 <= 0
r6: x6 <= 0
r7: x7 <= 0
r8: x8 <= 0
r9: x9 <= 0

end

CPLEX> read vrp0.lp
Problem 'vrp0.lp' read.
Read time =    0.03 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 14 rows and 14 columns.
All rows and columns eliminated.
Presolve time =    0.00 sec.

Dual simplex - Optimal: Objective = 4.5000000000e+01
Solution time =    0.00 sec. Iterations = 0 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
xa                  1.000000
xb                  1.000000
xc                  1.000000
xd                  1.000000
xe                  1.000000
All other variables in the range 1-14 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
c1                   11.000000
c2                   10.000000
c3                    9.000000
c4                    8.000000
c5                    7.000000
r1                   -18.000000
r2                    -8.000000
r3                   -16.000000
r4                    -8.000000
r5                    -7.000000
r6                   -13.000000
r8                    -6.000000
All other dual prices in the range 1-14 are 0.
```

Iteration 1

```
minimize
    11x1+10x2+11x3+ 9x4+ 9x5+11x6+ 8x7+ 9x8+ 7x9+11xa+10xb+ 9xc+ 8xd+ 7xe

subject to
c1:  +x1                                     +xa                                     >= 1
c2:  +x1  +x2  +x3                                     +xb                                     >= 1
c3:                                     +x3  +x4  +x5  +x6                                     +xc                                     >= 1
c4:  +x1  +x2  +x3  +x4                                     +x6  +x7  +x8                                     +xd                                     >= 1
c5:                                     +x5  +x6                                     +x8  +x9                                     +xe >= 1

\\ r1: x1 <= 0
r2: x2 <= 0
r3: x3 <= 0
r4: x4 <= 0
r5: x5 <= 0
r6: x6 <= 0
r7: x7 <= 0
r8: x8 <= 0
r9: x9 <= 0

end

CPLEX> read vrp1
Problem 'vrp1.lp' read.
Read time = 0.03 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 10 rows and 10 columns.
Reduced LP has 3 rows, 4 columns, and 6 nonzeros.
Presolve time = 0.00 sec.

Iteration log . . .
Iteration: 1 Dual objective = 27.000000

Dual simplex - Optimal: Objective = 2.7000000000e+01
Solution time = 0.00 sec. Iterations = 2 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
xc                  1.000000
xe                  1.000000
All other variables in the range 1-14 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
c1                   11.000000
c3                   9.000000
c5                   7.000000
r5                   -7.000000
r6                   -5.000000
All other dual prices in the range 1-13 are 0.
```

Iteration 2

```
minimize
    11x1+10x2+11x3+ 9x4+ 9x5+11x6+ 8x7+ 9x8+ 7x9+11xa+10xb+ 9xc+ 8xd+ 7xe

subject to
c1:  +x1                                +xa                                >= 1
c2:  +x1  +x2  +x3                                +xb                                >= 1
c3:                                +x3  +x4  +x5  +x6                                +xc                                >= 1
c4:  +x1  +x2  +x3  +x4                                +x6  +x7  +x8                                +xd                                >= 1
c5:                                +x5  +x6                                +x8  +x9                                +xe >= 1

\\ r1: x1 <= 0
r2: x2 <= 0
r3: x3 <= 0
r4: x4 <= 0
\\ r5: x5 <= 0
r6: x6 <= 0
r7: x7 <= 0
r8: x8 <= 0
r9: x9 <= 0

end

CPLEX> read vrp2
Problem 'vrp2.lp' read.
Read time =    0.02 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 7 rows and 7 columns.
Reduced LP has 5 rows, 7 columns, and 10 nonzeros.
Presolve time =    0.00 sec.

Iteration log . . .
Iteration:    1    Dual objective      =          11.000000

Dual simplex - Optimal:  Objective =  2.0000000000e+01
Solution time =    0.00 sec.  Iterations = 4 (0)

CPLEX> dis sol var -
Variable Name          Solution Value
x1                      1.000000
x5                      1.000000
All other variables in the range 1-14 are 0.
CPLEX> dis sol dua -
Constraint Name          Dual Price
c1                      11.000000
c3                      9.000000
All other dual prices in the range 1-12 are 0.
```